MATHS MANTHAN

• TUITION CLASSES •

e-mail: contact@mathsmanthan.co.in | Website: mathsmanthan.co.in

Subject: Mathematics | **PERIODIC TEST – 03 (SOLUTIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 12: Surface Areas and Volumes** | **MM:** 24

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, factoring common terms like πr before substituting $\pi = \frac{22}{7}$ or 3.14, showing intermediate arithmetic clearly, and boxing the final answer with proper physical units ($\text{cm}^2, \text{cm}^3, \text{m}^2, \text{m}^3$).
2. Partial step marks are allocated in brackets (**[0.5 Mark]**, **[1 Mark]**, **[1.5 Marks]**, **[2 Marks]**). Alternative valid methods receive full credit.
3. Total marks of printed questions Q1–Q11 $= 5 + 4 + 6 + 5 + 4 = 24$ Marks (matching section totals; source paper header prints MM: 30).

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective & MCQs

MATHS MANTHAN | $(5 \times 1 = 5$ Marks$)$

Q1. A circus tent is cylindrical to a height of 4 m and conical above it. If its diameter is 105 m and its slant height is 40 m, the total area of the canvas required in m^2 is: [1]

(A) 1760 (B) 2640 (C) 3960 (D) 7920

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(D) 7920 m^2** [0.5 Mark]

Given: $r = \frac{105}{2}$ m, $h_{\text{cyl}} = 4$ m, $l = 40$ m.

Formula / Theorem Used: Canvas Area $= 2\pi r h_{\text{cyl}} + \pi r l = \pi r(2h_{\text{cyl}} + l)$

$$\text{Area} = \frac{22}{7} \times \frac{105}{2} \times (8 + 40) = 11 \times 15 \times 48 = 165 \times 48 = \boxed{\mathbf{7920}\ \text{m}^2}$$

Q2. A sphere of radius 6 cm is dropped into a cylindrical vessel partly filled with water of radius 8 cm. When submerged completely, the water rises by: [1]

(A) 4.5 cm (B) 3.5 cm (C) 4.0 cm (D) 2.0 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(A) 4.5 cm** [0.5 Mark]

Given: Sphere radius $r = 6$ cm, cylinder radius $R = 8$ cm. Let water rise $= h$.

Formula / Theorem Used: $\pi R^2 h = \frac{4}{3}\pi r^3$

$$8^2 h = \frac{4}{3} \times 6^3 \implies 64h = \frac{4}{3} \times 216 = 288 \implies h = \frac{288}{64} = \frac{9}{2} = \boxed{\mathbf{4.5}\ \text{cm}}$$

Q3. If a cone is cut into two parts by a horizontal plane passing through the mid-point of its axis, the ratio of the volumes of the upper part and the entire cone is: [1]

(A) 1 : 2 (B) 1 : 4 (C) 1 : 6 (D) 1 : 8

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(D) 1 : 8** [0.5 Mark]

Given: By triangle similarity, the upper cone has height $\frac{h}{2}$ and radius $\frac{r}{2}$.

Formula / Theorem Used: $\frac{V_{\text{upper}}}{V_{\text{entire}}} = \frac{\frac{1}{3}\pi(r/2)^2(h/2)}{\frac{1}{3}\pi r^2 h} = \left(\frac{1}{2}\right)^3 = \boxed{\mathbf{1:8}}$

Q4. The height of a cone is 30 cm. A small cone is cut off at the top by a plane parallel to the base of volume $\frac{1}{27}$ of the cone. The height above the base at which the section is made is: [1]

(A) 10 cm (B) 15 cm (C) 20 cm (D) 25 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(C) 20 cm** [0.5 Mark]

Given: Total height $H = 30$ cm. Similar cones satisfy $\frac{V_{\text{small}}}{V_{\text{big}}} = \left(\frac{h}{H}\right)^3 = \frac{1}{27}$.

Formula / Theorem Used: $\frac{h}{30} = \sqrt[3]{\frac{1}{27}} = \frac{1}{3} \implies h = 10$ cm from vertex

Height above base $= 30 - 10 =$ **20 cm**. [0.5 Mark]

Q5. **Assertion (A):** Total surface area of a playing top formed by a cone mounted on a hemisphere is the sum of the curved surface area of the hemisphere and the curved surface area of the cone.
Reason (R): The top is obtained by fixing the flat circular surfaces of the hemisphere and cone together, making them internal. [1]

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(A)** [0.5 Mark]

Verification: Since the two circular bases coincide and become concealed inside the solid, only the two curved outer surfaces remain exposed ($\text{TSA} = \text{CSA}_{\text{hemi}} + \text{CSA}_{\text{cone}}$). Both (A) and (R) are true, and (R) explains (A).

SECTION B: Very Short Answer Questions MATHS MANTHAN | (2 × 2 = 4 Marks)

Q6. Two cubes have their volumes in the ratio 1 : 27. What is the ratio of their total surface areas? [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Finding Edge Ratio [1 Mark]

Let edges be a_1, a_2. Given $\frac{V_1}{V_2} = \frac{a_1^3}{a_2^3} = \frac{1}{27} \implies \frac{a_1}{a_2} = \sqrt[3]{\frac{1}{27}} = \frac{1}{3}$.

Formula / Theorem Used: $\frac{\text{TSA}_1}{\text{TSA}_2} = \frac{6a_1^2}{6a_2^2} = \left(\frac{a_1}{a_2}\right)^2$

Step 2: Squaring Ratio [1 Mark]

$$\frac{\text{TSA}_1}{\text{TSA}_2} = \left(\frac{1}{3}\right)^2 = \frac{1}{9} \implies \boxed{\mathbf{1:9}}$$

Q7. Two right circular cylinders of equal volumes have their heights in the ratio 1 : 2. What is the ratio of their radii? [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Equating Equal Volumes [1 Mark]

Given $V_1 = V_2$ and $\frac{h_1}{h_2} = \frac{1}{2}$.

Formula / Theorem Used: $\pi r_1^2 h_1 = \pi r_2^2 h_2 \implies \left(\frac{r_1}{r_2}\right)^2 = \frac{h_2}{h_1}$

Step 2: Taking Square Root [1 Mark]

$$\left(\frac{r_1}{r_2}\right)^2 = \frac{2}{1} \implies \frac{r_1}{r_2} = \frac{\sqrt{2}}{1} \implies \boxed{\mathbf{\sqrt{2}:1}}$$

Q8. A cylindrical container of radius 6 cm and height 15 cm is filled with ice-cream. The whole ice-cream is distributed to 10 children in equal cones with hemispherical tops. If the height of the conical portion is four times the radius of its base, find the radius of the ice-cream cone. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Volume Equations [1.5 Marks]

Container: $R = 6$ cm, $H = 15$ cm. Cone with hemispherical top: radius r, cone height $h = 4r$.

Formula / Theorem Used: $V_{\text{cyl}} = 10 \times (V_{\text{cone}} + V_{\text{hemi}}) = 10\left[\frac{1}{3}\pi r^2(4r) + \frac{2}{3}\pi r^3\right] = 10(2\pi r^3) = 20\pi r^3$

Step 2: Solving for r [1.5 Marks]

$$\pi(6)^2(15) = 20\pi r^3 \implies 36 \times 15 = 20r^3 \implies 540 = 20r^3 \implies r^3 = \frac{540}{20} = 27 \implies \boxed{\mathbf{r = 3}\text{ cm}}$$

Q9. A circus tent is cylindrical up to a height of 3 m and conical above it. If the diameter of the base is 105 m and the slant height of the conical part is 53 m, find the total canvas used in making the tent. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Dimensions & Formula [1 Mark]

Radius $r = \frac{105}{2}$ m, $h_{\text{cyl}} = 3$ m, $l = 53$ m.

Formula / Theorem Used: Total Canvas Area $= 2\pi r h_{\text{cyl}} + \pi r l = \pi r(2h_{\text{cyl}} + l)$

Step 2: Arithmetic [2 Marks]

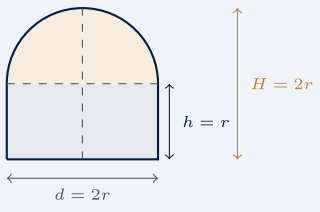
$$\text{Canvas} = \frac{22}{7} \times \frac{105}{2} \times (2 \times 3 + 53) = 11 \times 15 \times (6 + 53) = 165 \times 59 = \boxed{\mathbf{9735}\text{ m}^2}$$

Q10. A building is in the form of a cylinder surmounted by a hemispherical vaulted dome and contains $41\frac{19}{21}$ m^3 of air. If the internal diameter of the building is equal to its total height above the floor, find the height of the building. (Take $\pi = \frac{22}{7}$) [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Relating r and Cylinder Height h [1.5 Marks]

Let internal radius $= r$ m $\implies$ diameter $d = 2r$. Since total height $H = d = 2r$ and dome height $= r$, cylinder height $h = 2r - r = r$ m.



Formula / Theorem Used: $V_{\text{building}} = \pi r^2 h + \frac{2}{3}\pi r^3 = \pi r^3 + \frac{2}{3}\pi r^3 = \frac{5}{3}\pi r^3$

Step 2: Solving for r & Height H [3.5 Marks]

Given volume $= 41\frac{19}{21} = \frac{41\times21+19}{21} = \frac{861+19}{21} = \frac{880}{21}$ m^3.

$$\frac{5}{3} \times \frac{22}{7} \times r^3 = \frac{880}{21} \implies \frac{110}{21}r^3 = \frac{880}{21}$$

$$r^3 = \frac{880}{110} = 8 \implies r = 2\text{ m}$$

Height of building $H = 2r = 2 \times 2 = \boxed{\mathbf{4}\text{ m}}$.

Q11. Case Study: High-Precision Wooden Desk Pen Stand

A desk pen stand is manufactured from a block of wood shaped as a rectangular cuboid with four identical conical depressions drilled into the top surface to hold pens. Dimensions of the cuboid: Length = 15 cm, Breadth = 10 cm, Height = 3.5 cm. Dimensions of each conical depression: Radius $r = 0.5$ cm, Depth $h = 1.4$ cm. [4]

(i) Find the volume of the original cuboidal wooden block before depressions were drilled. [1]

(ii) Find the volume of each conical depression. [1]

(iii) Find the total volume of wood remaining in the finished pen stand. [1]

(iv) Find the total outer surface area of the cuboid. [1]

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Extracted Scenario Data: Cuboid: $l = 15$ cm, $b = 10$ cm, $H = 3.5$ cm. Cones (4): $r = 0.5$ cm, $h = 1.4$ cm.

Formula / Theorem Used: $V_{\text{cuboid}} = lbh$, $V_{\text{cone}} = \frac{1}{3}\pi r^2 h$, $\text{TSA}_{\text{cuboid}} = 2(lb + bH + Hl)$

Sub-part (i): Original Cuboid Volume [1 Mark]

$V_{\text{cuboid}} = 15 \times 10 \times 3.5 = 15 \times 35 = \boxed{\mathbf{525}\ \text{cm}^3}$.

Sub-part (ii): Volume of Each Conical Depression [1 Mark]

$V_1 = \frac{1}{3} \times \frac{22}{7} \times (0.5)^2 \times 1.4 = \frac{1}{3} \times 22 \times 0.25 \times 0.2 = \frac{1.1}{3} = \frac{11}{30} \approx \boxed{\mathbf{0.367}\ \text{cm}^3}$.

Sub-part (iii): Wood Remaining in Pen Stand [1 Mark]

$V_{4\ \text{cones}} = 4 \times \frac{1.1}{3} = \frac{4.4}{3} \approx 1.467\ \text{cm}^3 \implies V_{\text{wood}} = 525 - 1.467 = \boxed{\mathbf{523.53}\ \text{cm}^3}$.

Sub-part (iv): Total Outer Surface Area of Cuboid [1 Mark]

$\text{TSA} = 2(15 \times 10 + 10 \times 3.5 + 3.5 \times 15) = 2(150 + 35 + 52.5) = 2 \times 237.5 = \boxed{\mathbf{475}\ \text{cm}^2}$.

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★